

Linear Regression

Introduction

Linear regression is a **supervised learning algorithm** used for **predicting continuous values** based on one or more input features. It assumes a **linear relationship** between the dependent variable Y and one or more independent variables X .

Types of Linear Regression

1. **Simple Linear Regression** – When there is only **one** independent variable.
2. **Multiple Linear Regression** – When there are **two or more** independent variables.

1. Simple Linear Regression

Equation:

$$Y = mX + c + \epsilon$$

Where:

- Y = Dependent variable (Target)
- X = Independent variable (Feature)
- m = Slope (Coefficient)
- c = Intercept
- ϵ = Error term (random noise)

Example 1: Predicting House Prices

A real estate company wants to predict **house prices** based on the **size of the house**.

House Size (sq. ft.)	800	1000	1200	1500	1800
House Price (in \$1000s)	200	250	300	370	450

Using **linear regression**, we can estimate the best-fit line:

$$\text{House Price} = 50 + 0.2 \times (\text{Size})$$

So, if a house is **1400 sq. ft.**, the predicted price is:

$$50 + 0.2 \times 1400 = 330 \text{ (in \$1000s)} = \$330,000$$

2. Multiple Linear Regression

Equation:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n + \epsilon$$

Where:

- b_0 = Intercept
- $b_1, b_2, \dots, b_n$ = Coefficients of predictors
- $X_1, X_2, \dots, X_n$ = Independent variables
- ϵ = Error term

Example 2: Predicting Student Scores

A school wants to predict **student exam scores** based on **study hours** and **number of practice tests taken**.

Study Hours	2	3	5	7	9
Practice Tests	1	2	3	5	6
Exam Score (%)	50	55	65	80	85

Using multiple linear regression, we get:

$$\text{Score} = 20 + 5(\text{Study Hours}) + 3(\text{Practice Tests})$$

If a student studies **6 hours** and takes **4 practice tests**, the predicted score is:

$$20 + 5(6) + 3(4) = 20 + 30 + 12 = 62\%$$

Assumptions of Linear Regression

1. **Linearity:** The relationship between the dependent and independent variable(s) is linear.
2. **Independence:** The observations are independent.
3. **Homoscedasticity:** The variance of residuals is constant.
4. **Normality of Residuals:** The residuals (errors) should be normally distributed.
5. **No Multicollinearity (for Multiple Regression):** Independent variables should not be highly correlated.

Advantages of Linear Regression

- Simple and easy to interpret.
- Computationally efficient.
- Works well when the relationship is truly linear.

Disadvantages of Linear Regression

- Fails when the relationship is nonlinear.
- Sensitive to outliers.
- Assumes that independent variables are not correlated (multicollinearity issue).

Linear and Non-Linear Regression

Regression analysis is a fundamental statistical technique used to model relationships between a dependent variable and one or more independent variables. It helps in making predictions and understanding patterns in data.

1. Linear Regression

Linear regression assumes a **straight-line relationship** between the dependent variable Y and independent variable(s) X .

Equation of Linear Regression

For a single variable:

$$Y = mX + c + \epsilon$$

For multiple variables:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n + \epsilon$$

Where:

- Y = Dependent variable (target)
- X = Independent variable(s) (features)
- $m, b_1, b_2, \dots$ = Slopes (coefficients)
- c (or b_0) = Intercept
- ϵ = Error term (random noise)

Example 1: Predicting House Prices (Simple Linear Regression)

A real estate company wants to predict **house prices** based on **size (sq. ft.)**.

House Size (sq. ft.)	800	1000	1200	1500	1800
House Price (\$1000s)	200	250	300	370	450

The best-fit line from linear regression:

$$\text{Price} = 50 + 0.2 \times \text{Size}$$

If a house is **1400 sq. ft.**, the predicted price is:

$$50 + 0.2 \times 1400 = 330 \text{ (in \$1000s)} = 330,000$$

Example 2: Multiple Linear Regression (Predicting Sales)

A company wants to predict **sales revenue** based on **advertising spend on TV and social media**.

TV Ads (\$1000)	10	15	20	25
Social Media Ads (\$1000)	5	8	12	15
Sales Revenue (\$1000)	200	280	350	410

Using multiple linear regression, we get:

$$\text{Sales} = 50 + 5(\text{TV Ads}) + 3(\text{Social Media Ads})$$

If a company spends **\$18,000** on TV ads and **\$10,000** on social media ads, the predicted sales revenue is:

$$50 + 5(18) + 3(10) = 50 + 90 + 30 = 170 \text{ (in \$1000s)} = 170,000$$

Advantages of Linear Regression

- Easy to interpret.
- Computationally efficient.
- Works well if the relationship is **truly linear**.

Disadvantages of Linear Regression

- Assumes a linear relationship.
- Sensitive to outliers.
- Can struggle with **highly correlated features** (multicollinearity).

2. Non-Linear Regression

When the relationship between X and Y **cannot be represented by a straight line**, we use **non-linear regression**.

Types of Non-Linear Regression Models

1. **Polynomial Regression** – Fits a polynomial equation to the data.
2. **Exponential Regression** – Models exponential growth or decay.
3. **Logarithmic Regression** – Used when growth rate decreases over time.
4. **Power Regression** – Used in physics and engineering applications.
5. **Sigmoid and Logistic Regression** – Used for classification tasks.

Example 3: Polynomial Regression (Modeling Population Growth)

If a city's population over the years follows a **curved trend**, polynomial regression is used.

Year	2000	2005	2010	2015	2020
Population (in thousands)	50	80	130	210	320

A **quadratic** regression model:

$$\text{Population} = 10 + 2.5(\text{Year}) - 0.1(\text{Year}^2)$$

Example 4: Exponential Regression (COVID-19 Cases Growth)

During the early phase of the pandemic, the number of cases grew **exponentially**.

Days	1	5	10	15
Total Cases	100	500	2500	12500

An **exponential model**:

$$Y = 100e^{0.5X}$$

If $X = 20$, predicted cases:

$$Y = 100e^{0.5(20)} = 100e^{10} \approx 2.2 \times 10^6$$

Comparison: Linear vs. Non-Linear Regression

Feature	Linear Regression	Non-Linear Regression
Relationship	Straight-line	Curved (exponential, polynomial, etc.)
Equation	$Y = mX + c$	$Y = f(X)$ (complex functions)
Example	House prices vs. size	COVID-19 growth, Population increase
Complexity	Simple	More complex
Computational Cost	Low	Higher

3. When to Use Linear vs. Non-Linear Regression

✓ Use **Linear Regression** when:

- The relationship between variables is **linear**.
- There are **fewer** independent variables.
- Simplicity and interpretability are important.

✓ Use **Non-Linear Regression** when:

- The relationship is **curved or complex**.
- The model involves **exponential, logarithmic, or polynomial patterns**.
- More flexibility is required to capture data patterns.

Lines of Regression

1. Introduction to Regression Lines

Regression lines are used in **statistical modeling** to estimate the relationship between variables. When dealing with **two variables**, the regression lines help in predicting one variable based on the other.

If we have two variables, X (independent) and Y (dependent), we can define **two regression lines**:

1. **Regression of Y on X** → Predicts Y based on X .
2. **Regression of X on Y** → Predicts X based on Y .

2. Equations of Regression Lines

The two regression lines are given by:

1. **Regression Line of Y on X :**

$$Y = a + bX$$

- a = Intercept (value of Y when $X = 0$)
- b = Regression coefficient (b_{yx})
- X = Independent variable
- Y = Dependent variable

2. **Regression Line of X on Y :**

$$X = c + dY$$

- c = Intercept (value of X when $Y = 0$)
- d = Regression coefficient (b_{xy})

The **regression coefficients** are given by:

$$b_{yx} = \frac{r\sigma_Y}{\sigma_X}, \quad b_{xy} = \frac{r\sigma_X}{\sigma_Y}$$

where:

- r = **Correlation coefficient** between X and Y .
- σ_X, σ_Y = **Standard deviations** of X and Y .

The **product of regression coefficients** satisfies:

$$b_{yx} \times b_{xy} = r^2$$

If $r = 0$, the regression lines are perpendicular, meaning there is **no correlation** between X and Y .

3. Graphical Representation of Regression Lines

The two regression lines generally **intersect** at the mean point $(\bar{X}, \bar{Y})$. If the correlation is **perfect** (± 1), the two regression lines **coincide** into a single straight line.

Case 1: Perfect Positive Correlation ($r = 1$)

- The two regression lines coincide.
- The relationship is perfectly linear.

Case 2: No Correlation ($r = 0$)

- The regression lines are perpendicular to each other.
- There is no predictive relationship between X and Y .

Case 3: Moderate Correlation ($0 < r < 1$)

- The regression lines are distinct but close to each other.

4. Example 1: Regression of Height on Weight

A dataset records the **height (in cm)** and **weight (in kg)** of five individuals.

Person	A	B	C	D	E
Height (X) (cm)	160	165	170	175	180
Weight (Y) (kg)	55	58	60	63	67

Step 1: Calculate Means

$$\bar{X} = \frac{160 + 165 + 170 + 175 + 180}{5} = 170$$

$$\bar{Y} = \frac{55 + 58 + 60 + 63 + 67}{5} = 60.6$$

Step 2: Compute Regression Coefficients

Let the standard deviations be:

$$\sigma_X = 7.91, \quad \sigma_Y = 4.58, \quad r = 0.98$$

$$b_{yx} = \frac{r\sigma_Y}{\sigma_X} = \frac{0.98 \times 4.58}{7.91} = 0.57$$

$$b_{xy} = \frac{r\sigma_X}{\sigma_Y} = \frac{0.98 \times 7.91}{4.58} = 1.69$$

Step 3: Write Regression Equations

$$Y - 60.6 = 0.57(X - 170)$$

$$X - 170 = 1.69(Y - 60.6)$$

Using these, we can predict the weight for a given height or vice versa.

5. Example 2: Predicting Exam Scores

A study examines the **relationship between study hours (X) and exam scores (Y)**.

Study Hours (X)	2	3	5	6	8
Exam Score (Y)	50	55	70	75	85

From calculations, the regression equation is:

$$Y = 40 + 5X$$

If a student studies for **7 hours**, the predicted score is:

$$Y = 40 + 5(7) = 75$$

6. Properties of Regression Lines

1. **Regression lines always intersect at $(\bar{X}, \bar{Y})$.**
2. **If $r = \pm 1$, the two regression lines coincide.**
3. **If $r = 0$, the regression lines are perpendicular.**
4. **Regression coefficients satisfy $b_{yx} \times b_{xy} = r^2$.**
5. **Regression lines minimize the sum of squared errors (least squares method).**

Derivation of Regression Line of Y on X

1. Introduction

The **regression line of Y on X** represents the **best linear fit** for predicting Y given X. It is derived using the **least squares method**, which minimizes the sum of squared differences between actual and predicted values.

The equation of the regression line of Y on X is:

$$Y = a + bX$$

where:

- **a = Intercept** (value of Y when $X = 0$)
- **b = Regression coefficient** (b_{yx}), which measures the change in Y for a unit change in X.

2. Derivation of the Regression Equation of Y on X

We use the **least squares method** to derive the formula.

Step 1: Define the Error Term

For each observation (X_i, Y_i) , the predicted value is:

$$\hat{Y}_i = a + bX_i$$

The error e_i (residual) is:

$$e_i = Y_i - \hat{Y}_i = Y_i - (a + bX_i)$$

The least squares method minimizes the sum of squared errors:

$$S = \sum (Y_i - a - bX_i)^2$$

Step 2: Differentiate and Solve for a and b

To minimize S , we take partial derivatives with respect to a and b and set them to zero.

First equation (differentiating w.r.t. a):

$$\begin{aligned}\frac{\partial S}{\partial a} &= -2 \sum (Y_i - a - bX_i) = 0 \\ \sum Y_i &= na + b \sum X_i \\ a &= \frac{\sum Y_i - b \sum X_i}{n}\end{aligned}$$

Second equation (differentiating w.r.t. b):

$$\begin{aligned}\frac{\partial S}{\partial b} &= -2 \sum X_i(Y_i - a - bX_i) = 0 \\ \sum X_i Y_i &= a \sum X_i + b \sum X_i^2\end{aligned}$$

Step 3: Express a and b Using Means

Using the mean values:

$$\bar{X} = \frac{\sum X_i}{n}, \quad \bar{Y} = \frac{\sum Y_i}{n}$$

Rewriting a in terms of $\bar{X}$ and $\bar{Y}$:

$$a = \bar{Y} - b\bar{X}$$

Solving for b :

$$b = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

This gives the final regression equation:

$$Y - \bar{Y} = b(X - \bar{X})$$

where:

$$b = b_{yx} = \frac{r\sigma_Y}{\sigma_X}$$

Here,

- r = **Correlation coefficient**
- σ_X, σ_Y = **Standard deviations** of X and Y

Thus, the regression line of Y on X is:

$$Y = \bar{Y} + b_{yx}(X - \bar{X})$$

3. Example 1: Height and Weight

Given Data

Person	A	B	C	D	E
Height (X) (cm)	160	165	170	175	180
Weight (Y) (kg)	55	58	60	63	67

Step 1: Calculate Means

$$\bar{X} = \frac{160 + 165 + 170 + 175 + 180}{5} = 170$$

$$\bar{Y} = \frac{55 + 58 + 60 + 63 + 67}{5} = 60.6$$

Step 2: Compute Regression Coefficient b_{yx}

Let the correlation coefficient be $r = 0.98$, and standard deviations be:

$$\sigma_X = 7.91, \quad \sigma_Y = 4.58$$

$$b_{yx} = \frac{r\sigma_Y}{\sigma_X} = \frac{0.98 \times 4.58}{7.91} = 0.57$$

Step 3: Write Regression Equation

$$Y - 60.6 = 0.57(X - 170)$$

Expanding:

$$Y = 60.6 + 0.57X - 97$$

$$Y = 0.57X - 36.4$$

Step 4: Predict Weight for Height = 175 cm

$$Y = 0.57(175) - 36.4 = 63.35 \approx 63$$

Thus, the predicted weight for a 175 cm tall person is 63 kg.

4. Example 2: Study Hours vs Exam Score

Given Data

Study Hours (X)	2	3	5	6	8
Exam Score (Y)	50	55	70	75	85

Step 1: Compute Means

$$\bar{X} = \frac{2 + 3 + 5 + 6 + 8}{5} = 4.8, \quad \bar{Y} = \frac{50 + 55 + 70 + 75 + 85}{5} = 67$$

Step 2: Compute b_{yx}

Using formula:

$$b_{yx} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$$

$$b_{yx} = \frac{(2 - 4.8)(50 - 67) + (3 - 4.8)(55 - 67) + (5 - 4.8)(70 - 67) + (6 - 4.8)(75 - 67) + (8 - 4.8)(85 - 67)}{(2 - 4.8)^2 + (3 - 4.8)^2 + (5 - 4.8)^2 + (6 - 4.8)^2 + (8 - 4.8)^2}$$

$$b_{yx} = 5.1$$

Step 3: Write Regression Equation

$$Y - 67 = 5.1(X - 4.8)$$

$$Y = 67 + 5.1X - 24.48$$

$$Y = 5.1X + 42.52$$

Step 4: Predict Exam Score for 7 Hours Study

$$Y = 5.1(7) + 42.52 = 77.22 \approx 77$$

Thus, for **7 hours of study**, the predicted score is **77 marks**.

Derivation of Regression Line of X on Y

1. Introduction

The **regression line of X on Y** predicts the value of X for a given Y. It is derived using the **least squares method**, which minimizes the sum of squared differences between actual and predicted values of X.

The equation of the regression line of X on Y is:

$$X = a + bY$$

where:

- a = **Intercept** (value of X when $Y = 0$)
- b = **Regression coefficient** b_{xy} , which measures the change in X for a unit change in Y.

2. Derivation of the Regression Equation of X on Y

We use the **least squares method** to derive the formula.

Step 1: Define the Error Term

For each observation $((X_i, Y_i))$ the predicted value is:

$$\hat{X}_i = a + bY_i$$

The error e_i (residual) is:

$$e_i = X_i - \hat{X}_i = X_i - (a + bY_i)$$

The least squares method minimizes the sum of squared errors:

$$S = \sum (X_i - a - bY_i)^2$$

Step 2: Differentiate and Solve for a and b

To minimize S , we take partial derivatives with respect to a and b and set them to zero.

First equation (differentiating w.r.t. a):

$$\begin{aligned}\frac{\partial S}{\partial a} &= -2 \sum (X_i - a - bY_i) = 0 \\ \sum X_i &= na + b \sum Y_i \\ a &= \frac{\sum X_i - b \sum Y_i}{n}\end{aligned}$$

Second equation (differentiating w.r.t. b):

$$\frac{\partial S}{\partial b} = -2 \sum Y_i(X_i - a - bY_i) = 0$$

$$\sum X_i Y_i = a \sum Y_i + b \sum Y_i^2$$

Step 3: Express a and b Using Means

Using the mean values:

$$\bar{X} = \frac{\sum X_i}{n}, \quad \bar{Y} = \frac{\sum Y_i}{n}$$

Rewriting a in terms of $\bar{X}$ and $\bar{Y}$:

$$a = \bar{X} - b\bar{Y}$$

Solving for b :

$$b = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (Y_i - \bar{Y})^2}$$

This gives the final regression equation:

$$X - \bar{X} = b(Y - \bar{Y})$$

where:

$$b = b_{xy} = \frac{r\sigma_X}{\sigma_Y}$$

Here,

- r = **Correlation coefficient**
- σ_X, σ_Y = **Standard deviations** of X and Y

Thus, the regression line of X on Y is:

$$X = \bar{X} + b_{xy}(Y - \bar{Y})$$

3. Example 1: Height and Weight

Given Data

Person	A	B	C	D	E
Height (X) (cm)	160	165	170	175	180
Weight (Y) (kg)	55	58	60	63	67

Step 1: Calculate Means

$$\bar{X} = \frac{160 + 165 + 170 + 175 + 180}{5} = 170$$

$$\bar{Y} = \frac{55 + 58 + 60 + 63 + 67}{5} = 60.6$$

Step 2: Compute Regression Coefficient b_{xy}

Let the **correlation coefficient** be $r = 0.98$, and standard deviations be:

$$\sigma_X = 7.91, \quad \sigma_Y = 4.58$$

$$b_{xy} = \frac{r\sigma_X}{\sigma_Y} = \frac{0.98 \times 7.91}{4.58} = 1.69$$

Step 3: Write Regression Equation

$$X - 170 = 1.69(Y - 60.6)$$

Expanding:

$$X = 170 + 1.69Y - 102.41$$

$$X = 1.69Y + 67.59$$

Step 4: Predict Height for Weight = 65 kg

$$X = 1.69(65) + 67.59 = 177.44 \approx 177$$

Thus, the predicted height for a **65 kg** person is **177 cm**.

4. Example 2: Study Hours vs Exam Score

Given Data

Study Hours (X)	2	3	5	6	8
Exam Score (Y)	50	55	70	75	85

Step 1: Compute Means

$$\bar{X} = \frac{2 + 3 + 5 + 6 + 8}{5} = 4.8, \quad \bar{Y} = \frac{50 + 55 + 70 + 75 + 85}{5} = 67$$

Step 2: Compute b_{xy}

Using formula:

$$b_{xy} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(Y_i - \bar{Y})^2}$$

$$b_{xy} = 0.17$$

Step 3: Write Regression Equation

$$X - 4.8 = 0.17(Y - 67)$$

$$X = 0.17Y + 4.8 - 11.39$$

$$X = 0.17Y - 6.59$$

Step 4: Predict Study Hours for Exam Score = 80

$$X = 0.17(80) - 6.59 = 7.01 \approx 7$$

Thus, for **80 marks**, the predicted study hours are **7 hours**.

Angle Between Regression Lines

1. Introduction

The **angle between regression lines** is the angle formed by the two regression lines:

1. **Regression line of Y on X:** $Y - \bar{Y} = b_{yx}(X - \bar{X})$

2. **Regression line of X on Y:** $X - \bar{X} = b_{xy}(Y - \bar{Y})$

The angle (θ) between these two lines is given by the formula:

$$\tan \theta = \frac{|b_{yx} - b_{xy}|}{1 + b_{yx}b_{xy}}$$

where:

- b_{yx} = Regression coefficient of Y on X
- b_{xy} = Regression coefficient of X on Y

The angle **depends on the correlation coefficient** r , since:

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X}, \quad b_{xy} = r \frac{\sigma_X}{\sigma_Y}$$

2. Derivation of the Formula

From the regression equations:

1. **Slope of regression line of Y on X:** $m_1 = b_{yx} = r \frac{\sigma_Y}{\sigma_X}$

2. **Slope of regression line of X on Y:** $m_2 = \frac{1}{b_{yx}} = b_{xy} = r \frac{\sigma_X}{\sigma_Y}$

Using the formula for the **angle between two intersecting lines**:

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

Substituting $m_1 = b_{yx}$ and $m_2 = b_{xy}$:

$$\tan \theta = \frac{|b_{yx} - b_{xy}|}{1 + b_{yx}b_{xy}}$$

Thus, the angle between the regression lines depends on the regression coefficients.

3. Important Observations

1. If $r = \pm 1$ (perfect correlation):

- $b_{yx} = b_{xy}$, so

$$\tan \theta = 0 \Rightarrow \theta = 0^\circ$$

- The two lines **coincide**.

2. If $r = 0$ (no correlation):

- b_{yx} and b_{xy} are reciprocals.

- The lines are **perpendicular**, so

$$\theta = 90^\circ$$

3. For any other correlation, $0 < \theta < 90^\circ$.

4. Examples

Example 1: Marks vs Study Hours

Suppose we have the following data for **students' marks and study hours**:

Study Hours (X)	2	3	5	6	8
Marks (Y)	50	55	70	75	85

Let's assume:

- $\sigma_X = 2.5, \sigma_Y = 10$
- $r = 0.8$

Step 1: Compute Regression Coefficients

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X} = 0.8 \times \frac{10}{2.5} = 3.2, \quad b_{xy} = r \frac{\sigma_X}{\sigma_Y} = 0.8 \times \frac{2.5}{10} = 0.2$$

Step 2: Compute Angle

$$\tan \theta = \frac{|3.2 - 0.2|}{1 + (3.2 \times 0.2)}$$

$$\tan \theta = \frac{3}{1.64} = 1.83$$

$$\theta = \tan^{-1}(1.83) \approx 61.3^\circ$$

So, the angle between the regression lines is **61.3°**.

Example 2: Height vs Weight

Height (X) (cm)	160	165	170	175	180
Weight (Y) (kg)	55	58	60	63	67

Let:

- $\sigma_X = 7.91, \sigma_Y = 4.58$
- $r = 0.98$

Step 1: Compute Regression Coefficients

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X} = 0.98 \times \frac{4.58}{7.91} = 0.567$$

$$b_{xy} = r \frac{\sigma_X}{\sigma_Y} = 0.98 \times \frac{7.91}{4.58} = 1.69$$

Step 2: Compute Angle

$$\tan \theta = \frac{|0.567 - 1.69|}{1 + (0.567 \times 1.69)}$$

$$\tan \theta = \frac{1.123}{1.958} = 0.573$$

$$\theta = \tan^{-1}(0.573) \approx 29.8^\circ$$

So, the angle between the regression lines is **29.8°**.

5. Conclusion

- The **angle between regression lines** depends on the **correlation coefficient** r .
- If $r = \pm 1$, **lines coincide** ($\theta = 0^\circ$).
- If $r = 0$, **lines are perpendicular** ($\theta = 90^\circ$).
- Intermediate values of r lead to angles between 0° and 90° .

Coefficients of Regression

1. Introduction

The **regression coefficients** are the constants that measure the relationship between two variables in a regression equation. They indicate the **rate of change** of one variable with respect to another.

For **two variables** X and Y , there are **two regression coefficients**:

1. **Regression coefficient of Y on X (b_{yx})**
 - Measures the change in Y for a unit change in X .
 - Used in the regression equation: $Y - \bar{Y} = b_{yx}(X - \bar{X})$
2. **Regression coefficient of X on Y (b_{xy})**
 - Measures the change in X for a unit change in Y .
 - Used in the regression equation: $X - \bar{X} = b_{xy}(Y - \bar{Y})$

The **correlation coefficient** r is related to the regression coefficients by:

$$r^2 = b_{yx} \times b_{xy}$$

Also, the **geometric mean** of the two regression coefficients is equal to the correlation coefficient:

$$r = \sqrt{b_{yx} \times b_{xy}}$$

2. Properties of Regression Coefficients

1. **The regression coefficients have the same sign as the correlation coefficient (r):**
 - If r is positive, both b_{yx} and b_{xy} are positive.
 - If r is negative, both b_{yx} and b_{xy} are negative.
2. **The product of the two regression coefficients is equal to the square of the correlation coefficient:**

$$b_{yx} \times b_{xy} = r^2$$

3. **If one of the regression coefficients is greater than 1, the other must be less than 1.**
4. **The two regression lines intersect at the mean point $(\bar{X}, \bar{Y})$.**

3. Formulas for Regression Coefficients

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X}$$

$$b_{xy} = r \frac{\sigma_X}{\sigma_Y}$$

Where:

- b_{yx} = Regression coefficient of Y on X
- b_{xy} = Regression coefficient of X on Y
- r = Correlation coefficient
- σ_X, σ_Y = Standard deviations of X and Y

4. Examples

Example 1: Student Study Hours vs Marks

Let's consider the following dataset:

Study Hours (X)	2	3	5	6	8
Marks (Y)	50	55	70	75	85

Step 1: Calculate Mean and Standard Deviations

Assume:

- $\bar{X} = 4.8, \bar{Y} = 67$
- $\sigma_X = 2.5, \sigma_Y = 10$
- Correlation coefficient: $r = 0.8$

Step 2: Compute Regression Coefficients

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X} = 0.8 \times \frac{10}{2.5} = 3.2$$

$$b_{xy} = r \frac{\sigma_X}{\sigma_Y} = 0.8 \times \frac{2.5}{10} = 0.2$$

Step 3: Regression Equations

1. Equation of Y on X :

$$Y - 67 = 3.2(X - 4.8)$$

$$Y = 3.2X + 51.64$$

2. Equation of X on Y :

$$X - 4.8 = 0.2(Y - 67)$$

$$X = 0.2Y + 1.4$$

Step 4: Verification

$$b_{yx} \times b_{xy} = 3.2 \times 0.2 = 0.64$$

$$r^2 = (0.8)^2 = 0.64$$

So, the property $b_{yx} \times b_{xy} = r^2$ holds.

Example 2: Height vs Weight

Height (X) (cm)	160	165	170	175	180
Weight (Y) (kg)	55	58	60	63	67

Assume:

- $\bar{X} = 170, \bar{Y} = 60.6$

- $\sigma_X = 7.91, \sigma_Y = 4.58$
- $r = 0.98$

Step 1: Compute Regression Coefficients

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X} = 0.98 \times \frac{4.58}{7.91} = 0.567$$

$$b_{xy} = r \frac{\sigma_X}{\sigma_Y} = 0.98 \times \frac{7.91}{4.58} = 1.69$$

Step 2: Regression Equations

1. Equation of Y on X:

$$Y - 60.6 = 0.567(X - 170)$$

$$Y = 0.567X - 35.79$$

2. Equation of X on Y:

$$X - 170 = 1.69(Y - 60.6)$$

$$X = 1.69Y + 67.86$$

Step 3: Verification

$$b_{yx} \times b_{xy} = 0.567 \times 1.69 = 0.958$$

$$r^2 = (0.98)^2 = 0.9604$$

Since the values are close, the regression coefficients satisfy the fundamental property.

Theorems on Regression Coefficients

1. Introduction

Regression coefficients are crucial in regression analysis, as they quantify the relationship between dependent and independent variables. Several **theorems** govern their properties and behavior.

2. Theorems on Regression Coefficients

Theorem 1: The Two Regression Coefficients Have the Same Sign

- If one regression coefficient is **positive**, the other must also be **positive**.
- If one regression coefficient is **negative**, the other must also be **negative**.

Proof:

The two regression coefficients are related to the correlation coefficient r :

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X}, \quad b_{xy} = r \frac{\sigma_X}{\sigma_Y}$$

Since r determines the sign, both b_{yx} and b_{xy} will always have the same sign as r .

Example 1:

Let $r = 0.85, \sigma_X = 3\sigma$, and $\sigma_Y = 4$.

$$b_{yx} = 0.85 \times \frac{4}{3} = 1.13, \quad b_{xy} = 0.85 \times \frac{3}{4} = 0.64$$

Both coefficients are **positive**, confirming the theorem.

Theorem 2: The Product of the Regression Coefficients is Equal to the Square of the Correlation Coefficient

$$b_{yx} \times b_{xy} = r^2$$

Proof:

Using the formulas:

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X}, \quad b_{xy} = r \frac{\sigma_X}{\sigma_Y}$$

Multiplying both,

$$\begin{aligned} b_{yx} \times b_{xy} &= \left(r \frac{\sigma_Y}{\sigma_X} \right) \times \left(r \frac{\sigma_X}{\sigma_Y} \right) \\ &= r^2 \times \frac{\sigma_Y}{\sigma_X} \times \frac{\sigma_X}{\sigma_Y} \\ &= r^2 \end{aligned}$$

Example 2:

If $r = 0.6$, $\sigma_X = 5$, and $\sigma_Y = 10$:

$$b_{yx} = 0.6 \times \frac{10}{5} = 1.2$$

$$b_{xy} = 0.6 \times \frac{5}{10} = 0.3$$

$$b_{yx} \times b_{xy} = 1.2 \times 0.3 = 0.36$$

$$r^2 = (0.6)^2 = 0.36$$

Thus, the theorem holds.

Theorem 3: If One Regression Coefficient is Greater than 1, the Other Must be Less than 1

Proof:

From the equation:

$$b_{yx} \times b_{xy} = r^2$$

Since $r^2 \leq 1$, if $b_{yx} > 1$, then $b_{xy} < 1$ to satisfy the equation.

Example 3:

Let $r = 0.5$, $\sigma_X = 4$, and $\sigma_Y = 2$:

$$b_{yx} = 0.5 \times \frac{2}{4} = 0.25$$

$$b_{xy} = 0.5 \times \frac{4}{2} = 1.0$$

Since $b_{yx} = 0.25 (< 1)$ and $b_{xy} = 1.0 (> 1)$, the theorem holds.

Theorem 4: The Regression Coefficients are Independent of Change of Origin but Not of Scale

- If we **change the origin** of X and Y (i.e., shifting values by adding/subtracting a constant), **the regression coefficients remain unchanged.**

- If we **change the scale** (i.e., multiplying by a factor), **the regression coefficients change**.

Proof:

Let new variables be:

$$X' = \frac{X - a}{h}, \quad Y' = \frac{Y - b}{k}$$

The regression coefficients become:

$$b_{y'x'} = b_{yx} \times \frac{k}{h}$$

Since h and k represent scale factors, the regression coefficients change with scale but not with origin.

Example 4:

Suppose we have:

X	10	20	30
Y	20	30	40

Let's shift X by 5 ($X' = X - 5$) and Y by 10 ($Y' = Y - 10$).

b_{yx} remains the same

Now, if we scale X by 2 ($X' = X/2$) and Y by 3 ($Y' = Y/3$),

b_{yx} changes by $\frac{3}{2}$

Thus, the theorem is verified.

Theorem 5: The Angle Between Two Regression Lines is Given by

$$\tan \theta = \frac{|\sigma_Y b_{yx} - \sigma_X b_{xy}|}{\sigma_Y + \sigma_X}$$

Example 5:

$$\begin{aligned} \text{For } b_{yx} = 1.2 \text{ and } b_{xy} = 0.8, \text{ with } \sigma_X = 5 \text{ and } \sigma_Y = 4, \\ \tan \theta = \frac{|(4 \times 1.2) - (5 \times 0.8)|}{4 + 5} = \frac{|4.8 - 4|}{9} = \frac{0.8}{9} = 0.089 \\ \theta = \tan^{-1}(0.089) \approx 5.1^\circ \end{aligned}$$

3. Summary of Theorems

Theorem	Statement
1	The two regression coefficients have the same sign.
2	The product of regression coefficients equals r^2 .
3	If one regression coefficient is >1 , the other must be <1 .
4	Regression coefficients are independent of change in origin but not scale.
5	The angle between regression lines by

Theorem	Statement
	$\tan \theta = \frac{ \sigma_Y b_{yx} - \sigma_X b_{xy} }{\sigma_Y + \sigma_X}$

Properties of Regression Coefficients

1. Introduction

Regression coefficients quantify the relationship between an independent variable (X) and a dependent variable (Y). They describe how much Y changes for a unit change in X . Several important **properties** govern regression coefficients, which help in understanding their behavior.

2. Properties of Regression Coefficients

Property 1: The Two Regression Coefficients Have the Same Sign

- If b_{yx} (regression coefficient of Y on X) is **positive**, then b_{xy} (regression coefficient of X on Y) is also positive.
- If b_{yx} is **negative**, then b_{xy} is also negative.

Proof:

Since both regression coefficients are related to the **correlation coefficient** (r):

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X}, \quad b_{xy} = r \frac{\sigma_X}{\sigma_Y}$$

Since r determines the sign, both coefficients will always have the **same sign as r** .

Example 1:

Let $r = 0.75$, $\sigma_X = 4$, and $\sigma_Y = 5$.

$$b_{yx} = 0.75 \times \frac{5}{4} = 0.9375, \quad b_{xy} = 0.75 \times \frac{4}{5} = 0.6$$

Since both are **positive**, the theorem holds.

Property 2: The Product of the Two Regression Coefficients is Equal to the Square of the Correlation Coefficient

$$b_{yx} \times b_{xy} = r^2$$

Proof:

Using the formulas:

$$b_{yx} = r \frac{\sigma_Y}{\sigma_X}, \quad b_{xy} = r \frac{\sigma_X}{\sigma_Y}$$

Multiplying both,

$$\begin{aligned} b_{yx} \times b_{xy} &= \left(r \frac{\sigma_Y}{\sigma_X} \right) \times \left(r \frac{\sigma_X}{\sigma_Y} \right) \\ &= r^2 \times \frac{\sigma_Y}{\sigma_X} \times \frac{\sigma_X}{\sigma_Y} \\ &= r^2 \end{aligned}$$

Example 2:

If $r = 0.8$, $\sigma_X = 6$, and $\sigma_Y = 3$:

$$b_{yx} = 0.8 \times \frac{3}{6} = 0.4, \quad b_{xy} = 0.8 \times \frac{6}{3} = 1.6$$

$$b_{yx} \times b_{xy} = 0.4 \times 1.6 = 0.64$$

Since $r^2 = (0.8)^2 = 0.64$, the theorem holds.

Property 3: If One Regression Coefficient is Greater Than 1, the Other Must Be Less Than 1

Since:

$$b_{yx} \times b_{xy} = r^2 \leq 1$$

If $b_{yx} > 1$, then b_{xy} must be **less than 1** for the product to be less than or equal to 1.

Example 3:

If $b_{yx} = 1.5$, then:

$$b_{xy} = \frac{r^2}{b_{yx}}$$

If $r = 0.8$, then:

$$b_{xy} = \frac{0.64}{1.5} = 0.426$$

Thus, one coefficient is greater than 1, and the other is less than 1.

Property 4: The Regression Coefficients are Independent of the Change of Origin but Not of Scale

- If we **shift** the origin by adding or subtracting a constant to X or Y , the regression coefficients remain unchanged.
- If we **change the scale** by multiplying by a factor, the regression coefficients **change**.

Proof:

Let new variables be:

$$X' = \frac{X - a}{h}, \quad Y' = \frac{Y - b}{k}$$

The regression coefficient transforms as:

$$b_{y'x'} = b_{yx} \times \frac{k}{h}$$

Since h and k represent **scale factors**, the regression coefficients **change with scale but not with origin**.

Example 4:

Suppose we have:

X	10	20	30
Y	15	25	35

Let's shift X by 5 ($X' = X - 5$) and Y by 10 ($Y' = Y - 10$).

b_{yx} remains the same

Now, if we scale X by 2 ($X' = X/2$) and Y by 3 ($Y' = Y/3$),

$$b_{yx} \text{ changes by } \frac{3}{2}$$

Thus, the theorem is verified.

Property 5: The Angle Between the Two Regression Lines is Given by

$$\tan \theta = \frac{|\sigma_Y b_{yx} - \sigma_X b_{xy}|}{\sigma_Y + \sigma_X}$$

Example 5:

For $b_{yx} = 1.2$ and $b_{xy} = 0.8$, with $\sigma_X = 5$ and $\sigma_Y = 4$,

$$\tan \theta = \frac{|(4 \times 1.2) - (5 \times 0.8)|}{4 + 5} = \frac{|4.8 - 4|}{9} = \frac{0.8}{9} = 0.089$$

$$\theta = \tan^{-1}(0.089) \approx 5.1^\circ$$

3. Properties in Short

Property	Statement
1	The two regression coefficients have the same sign.
2	The product of regression coefficients equals r^2 .
3	If one regression coefficient is >1 , the other must be <1 .
4	Regression coefficients are independent of change in origin but not scale.
5	The angle between regression lines